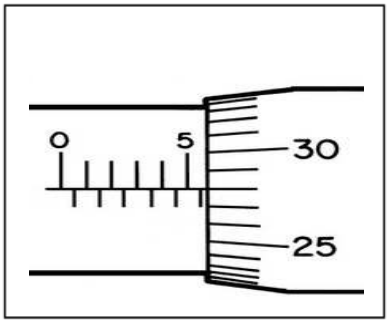
In measurement experiment, In an experiment to measure the density of a small metallic sphere, the diameter of the sphere (d) was measured by micrometer shown in the figure, and the mass of the sphere M was M ± ΔM = (2.1 ± 0.05) g. calculate the density of the sphere, ± Δ

A. 20792 ± 602.97
B. 24893 ± 652.98
C. 30552 ± 56.77 *D.* 10762 ± 302.96 *E.* Non of above



In measurement experiment, read the scale of the Vernier Caliper:

A. 2.4 cm
B. 1.6 cm
C. 11.9 cm
D. *5.1 cm*
E. *Non of above*

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Experimental value of gravity is found to be 10.07 m/s^2. Accepted value of g is 9.81m/s^2. Calculate the percent error (%)

A. *0.265 %*
B. *265 %*
C. *26.5 %*
D. *2.65 %*
E. *Non of above*

In force table experiment, find the resultant force of the following two forces: F1 = 1.96 N (weight of 200 gram) at 60^0 from positive x axis. F2 = 2.94 N (weight of 300 gram) at 140^0 from the positive x axis. Calculate the magnitude of resultant force

A. *4.8 N*
B. *3.8 N*
C. *2.8 N*
D. *1.8 N*
E. *Non of above*

In force table experiment, find the resultant force of the following two forces: F1 = 1.96 N (weight of 200 gram) at 60° from positive x axis. F2 = 2.94 N (weight of 300 gram) at 140° from the positive x axis. Calculate the direction of resultant force.

A. *110°*
B. *120°*
C. *130°*
D. *140°*
E. *Non of above*

In friction force experiment, for the inclined surface, if $_k = _s$=0.3; if the mass slides on the surface with constant velocity. Find the angle of the inclined surface?

A. *11°*
B. *12°*
C. *17°*
D. *14°*
E. *Non of above*

In uniform circular motion experiment, a ball of mass 0.5 kg is attached to the end of a string 1.5 m long and whirled in a horizontal plane at constant speed. If the cord can withstand a maximum tension of 50 N, what is the maximum speed of the ball before the string breaks?

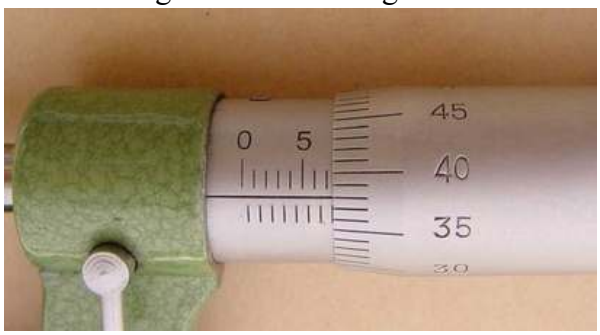
A. *18.3*
B. *12.2*
C. *25*
D. *7.5*
E. *Non of above*

A cylinder has a radius of R ± ΔR = (2.00 ± 0.05) cm and a length of L ± ΔL= (20.0 ± 0.1) cm. Calculate the volume of the cylinder V ± ΔV

V **V**= $2.51x10^{-4}$±$1.38x10^{-5}$
A. $2.51x10^{-3}$±$1.38x10^{-3}$
B. $2.51x10^{-4}$±$1.38x10^{-4}$
C. $2.51x10^{-3}$±$1.38x10^{-5}$
D. $2.51x10^{-4}$±$1.38x10^{-5}$
E. *Non of above*

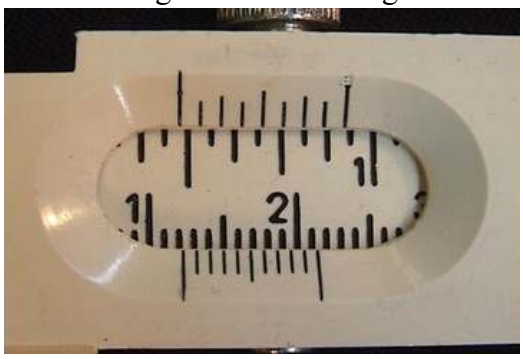
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What are the readings of the following instruments?



A. 7.38 ± 0.05 mm
B. 7.88 ± 0.005 mm
C. 7.88 ± 0.05 mm
D. 7.38 ± 0.005 mm
E. *Non of above*

What are the readings of the following instruments?



A. 1.23 ± 0.05cm
B. 1.23 ± 0.005cm
C. 1.23 ± 0.5cm 1.20
D. ± 0.05cm *Non*
E. *ofabove*

Round the number 62.5347 to four significant figures

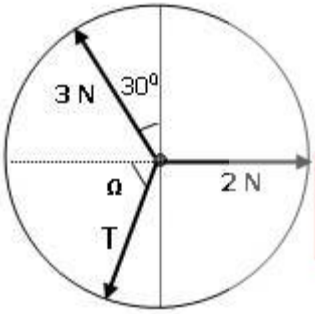
A. 63.00
B. 62.50
C. 62.54
D. 62.53
E. *Non ofabove*

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Round the number 3.78721 to three significant figures

A. 3.78
B. 3.79
C. 3.80
D. 4.00
E. *Non ofabove*

For the experimentalsetup shown, consider a body located at the origin, the body is in equilibrium, if T (inN), and Ω (in degree) are respectively:



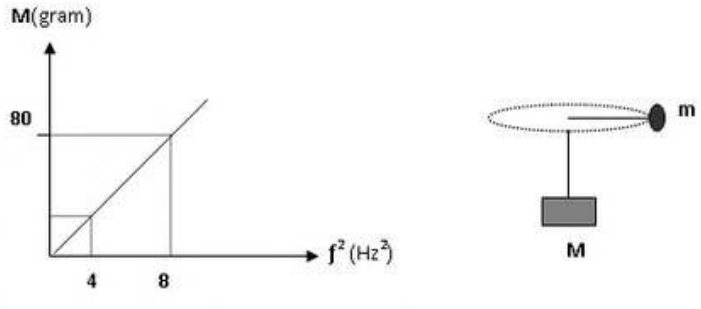
A. *3.36, 68.9*
B. *3.36, 79.1*
C. *2.64, 79.1*
D. *2.46, 79.1*
E. *Non of above*

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Which ofthefollowingmeasurements represents the reading of the caliber?

A. 2mm
B. 8.0 cm
C. 1.50 cm
D. 2.01mm
E. *Non ofabove*

In the circular motion experiment, a student studied the relation between the hanging mass (M) and the square of the frequency (f^2) for rotational body (m) using the radius (R=0.3 m). A plotted data are shown in the figure bellow. The mass (m) of the rotational body is:

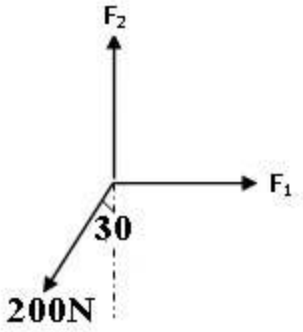


A. 12.7 gram
B. 4.22 gram
C. 6.34 gram
D. 8.44 gram
E. *Non ofabove*

An object is movingwithaconstant speed in a circular path of radius r. If its frequency of revolution is (3n), expressits centripetal acceleration in terms of n and r.

A. 36 $^2n^2r$
B. 3 $^2n^2r$
C. 9 $^2n^2r$
D. 4 $^2n^2r$
E. *Non ofabove*

The set of forces shown in the following figure are in equilibrium, the magnitudes of F_1 and F_2 (in N) are respectively given by:

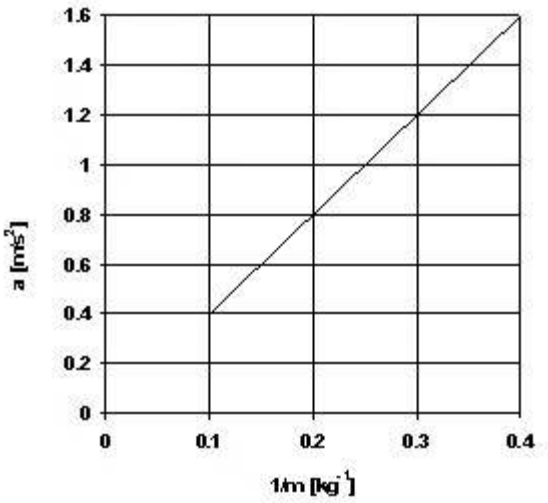


A. (100,173)
B. (173,100)
C. (-100,-173)
D. (-173,-100)
E. *Non ofabove*

Which of the following measurements represents the reading of the micrometer?

A. 9.25 mm
B. 8.0 cm
C. 1.50 cm
D. 2 mm
E. *Non of above*

In the Newton's second law experiment, the relation between the acceleration (a) and the total mass (m) with a constant force (F) is shown in Figure. The applied force on the system (in N) is:



A. *4 N*
B. *12 N*
C. *16 N*
D. *8 N*
E. *Non of above*

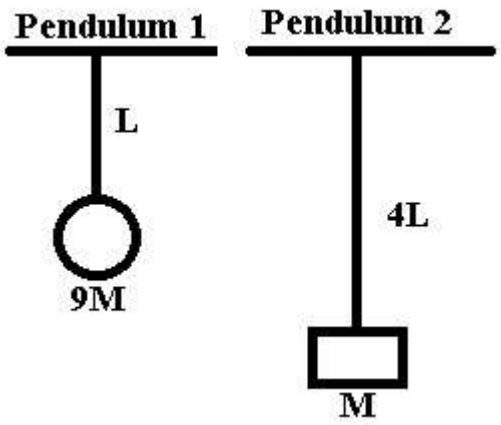
A simple pendulum has a simple harmonic motion if:

A. The string is very long
B. The angle gives sin ≈ .
C. The mass to length ratio is small
D. The mass to length ratio is large.
E. *Non of above*

In spring constant experiment, if the hanging mass 80g and the elongation in the spring is 4cm, the spring constant (N/m) is:

A. *20*
B. *40*
C. *5*
D. *10*
E. *Non of above*

In the simple pendulum experiment, the following figures show two pendulums. If the periodic time is T_1 for the Pendulum (1), and the periodic time is T_2 for the pendulum (2), then the relation between T_1 and T_2 is



A. $T_2=1.73T_1$
B. $T_2=2T_1$
C. $T_2 = 2.24T_1$ D.
$T_2 = 2.45T_1$ E.
Non of above

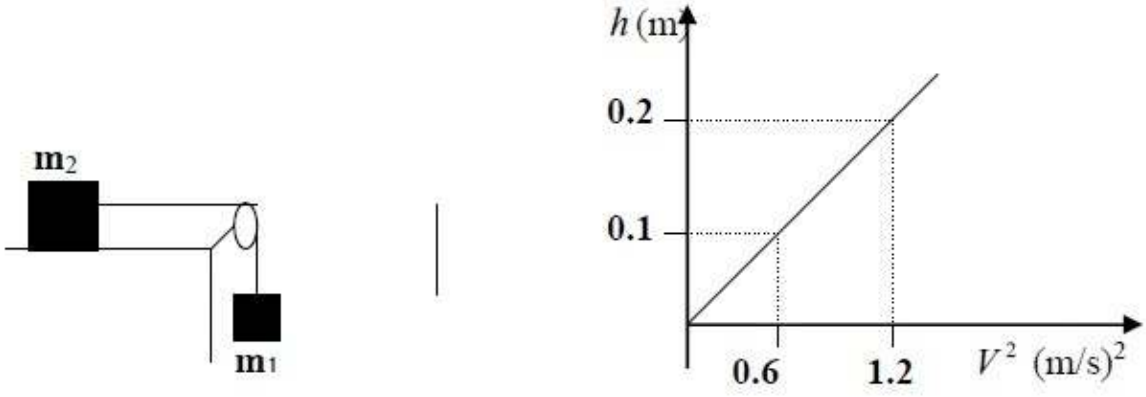
If F is the force, x the displacement, and k a spring constant, for simple harmonic motion, F is given by:

A. F=kx
B. F=-k/x
C. $F = -kx^2$
D. F = -kx
E. Non of above

In the simple harmonic oscillator, a mass of **(40g)** is hanged in a spring, if the extension of the spring is (**4 cm**) and if the mass drawn to (3cm) from equilibrium point and oscillated **25 times**, the total time of an oscillation of 20 times is (in sec):

A. 11.91
B. 9.93
C. 7.95
D. 5.96
E. Non of above

In the conservation of energy experiment, a student obtained data plotted in the following figure. If the masses $\mathbf{m_1=0.15\ kg}$ and $\mathbf{m_2=0.2\ kg}$, the acceleration of gravity (in m/s^2) is:



A. 11
B. 9
C. 13
D. 7
E. Non of above

In the elastic collisions, the linear momentum is conserved, while the kinetic energy is not:
A. False
B. True

In uniform circular motion, the peed is constant but the velocity is variable
A. False
B. True

In vector experiment, the mass hangers do not affect on the equilibrium state
C. False
D. True

The using of the air track in Newton's second law experiment is to decrease the friction effects
C. False
D. True

The sum of the change in kinetic and potential energy is always positive
A. False
B. True

The best equipment to measure the thickness of the paper is the Vernier caliper
A. False
B. True

We can conclude from the frictional force that μk is smaller than μ_s

A. False
B. True

The micrometer is more accurate than the vernier caliper

A. False
B. True

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